Mid- to Late-Holocene Landscape Evolution, Forest Dynamics, and Fluvial Regimes Across the Afro-Eurasian Arid-Temperate Belt (3500–300 BC)

Macro-Climatic Forcing and Geomorphic Feedbacks Across the Afro-Eurasian Loop

Between 3500 BC and 300 BC (ca. 5450–2250 cal yr BP), the contiguous landmass stretching from the Nile Valley through Anatolia, across the Iranian and Central Asian interior to the Kashmir Himalayas, and back through Mesopotamia, the Levant, and the Arabian Peninsula underwent a profound geo-ecological reorganization[1]. This 3,200-year span encompasses the transition from the terminal Mid-Holocene Climatic Optimum into the Late Holocene aridification regime, an era defined by the progressive decoupling of human socio-environmental systems from purely climatic forcing and the emergence of anthropogenically engineered landscapes[1].

At the macro-climatic scale, environmental transformations across this geographic loop were governed by the shifting equilibrium between two primary atmospheric circulation systems interacting with major orographic barriers—specifically the Taurus, Zagros, Hindu Kush, Pamir, Pir Panjal, and Hajar mountain ranges—which dictated regional rain shadows, alluvial fan formation, and montane forest refugia[6]. The first of these circulation systems, the subtropical monsoonal regime, comprised the Indian Summer Monsoon (ISM) and the African Monsoon,